

No. of Functional Features Included in the Solution

Date	08/07/2026
Team ID	
Project Name	HDI Predictor
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features implemented in the HDI Predictor — Policy Reviewer solution, built as a React + TypeScript single-page application with an Express backend. Each feature is described with its purpose, implementation status, and the source module in which it resides. The solution computes a country's Human Development Index using the UNDP geometric mean formula, and extends this with a policy intervention simulator, global country benchmarking, and an AI-powered policy advisory engine.

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	Live HDI Predictor with Sliders	Interactive sliders for life expectancy, mean years of schooling, expected years of schooling, and GNI per capita, computing the HDI score in real time using the UNDP geometric mean formula.	HdiPredictor.tsx / App.tsx (calculateHdiFromIndicators)	Done	1
2	Scenario & Country Presets	Four pre-built development scenarios (Very High, High, Medium, Low) plus a searchable list of real country presets that auto-fill all four indicators.	data.ts (SCENARIOS, COUNTRIES) / HdiPredictor.tsx	Done	0.5
3	Policy Intervention Simulator	A checklist of 6 real-world policy interventions (health, education, income) that apply boost multipliers to the baseline indicators, letting users simulate 'what-if' improvement scenarios.	data.ts (POLICY_INTERVENTIONS) / App.tsx (getSimulatedValues)	Done	1
4	Baseline vs Simulated HDI Comparison	Displays both the original baseline HDI score and the simulated HDI score side by side, showing the projected impact of selected policy interventions.	App.tsx (baselineHdi, simulatedHdi) / HdiPredictor.tsx	Done	0.5
5	AI Policy Advisor (Gemini Integration)	Sends the current indicator values and HDI category to a backend API endpoint which consults the Gemini AI model to generate a personalised policy advisory report.	server.ts (/api/advisor) / App.tsx (handleConsultAiAdvisor)	Done	1
6	Global Country Comparison Dashboard	A dedicated comparison view that benchmarks the user's custom country profile against real countries and global average indicators across all three HDI dimensions.	GlobalComparison.tsx	Done	0.5
7	HDI Report Generator	Generates a structured summary report of the current indicator inputs, sub-index breakdown, and final HDI classification for the selected or custom country profile.	HdiReport.tsx	Done	0.5
8	About HDI & About Project Pages	Educational pages explaining the three HDI pillars (health, education, income) and the UNDP methodology, plus a project information page describing the platform's purpose and data sources.	AboutHdi.tsx / AboutProject.tsx	Done	0.5

Feature Summary:

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	8
Core / Must-Have Features	5 (HDI Predictor, Scenario Presets, Policy Simulator, Baseline vs Simulated Comparison, HDI Report)
Additional / Nice-to-Have Features	3 (AI Policy Advisor, Global Comparison Dashboard, About HDI / About Project pages)
Features Tested & Verified	8

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	4	Interactive sliders, dark/light mode toggle, scenario & country preset selector, tabbed navigation (Home, Predictor, Comparison, Report, About)
2	Backend / Logic	3	calculateHdiFromIndicators() geometric mean engine, getSimulatedValues() policy boost calculator, HDI tier classification logic
3	Database / Storage	1	Static in-memory datasets: SCENARIOS, POLICY_INTERVENTIONS, COUNTRIES, GLOBAL_AVERAGES (data.ts)
4	API / Integration	1	Express server (server.ts) exposing POST /api/advisor endpoint, integrating the Gemini API for AI-generated policy recommendations
5	Security / Authentication	1	Environment variable-based API key management (.env.local) for the Gemini API; no user authentication required as the tool is public/stateless